

Approach Description: Team MorData

- **Data Collection**

Given Data: 570 Manual Post-Event Annotations & Building's footprint

External Datasets : A suite of diverse datasets, each serving a specific purpose in the training and enhancement of our models. Each external dataset will be detailed further in an accompanying readme file, providing in-depth insights into their composition.

- **Annotation Step**

We refined our dataset by segmenting high-quality TIFF images and filtering out any with cloud coverage, ensuring clarity for model training. By employing a pre-trained YOLO model on a dataset with diverse building conditions, we accelerated the annotation, applying a threshold of 10 bounding boxes per image for consistent quality. Despite challenges in classifying ambiguous structures, our collaborative efforts on the CVAT platform ensured accurate annotations of 570 images. Which gives 10929 undamagedresidentialbuilding 1369 damagedresidentialbuilding 1480 undamagedcommercialbuilding and 271 damagedcommercialbuilding

- **Approach Overview: 2 STAGES PREDICTION** (*figure 1*)

Motivations: Upon acquiring the data and initial interaction with the given base model, YOLO, we identified the classification task as the more challenging component of our project. The YOLOv8 model, even with relatively limited resources, demonstrated a high capability to detect a vast number of buildings. This observation led us to devise a two-stage predictive framework.

Stage 1- Building Detector - : Buildings possess distinctive features that differentiate them within their surroundings, regardless of their condition (e.g., damaged) or type (e.g., commercial). Consequently, we delineated this as a separate task, beginning with models designed for box and polygon proposals (we will explain later why polygons)

Stage - Building Classifier - : Following the identification of buildings within larger images, we isolate the structures, including a margin of context around each, for further analysis. This isolated image is then processed through two distinct classifiers: one for detecting damaged buildings and another for identifying commercial buildings

- **Important Points & Ideas**

Why 2 classifiers? Due to its ability to incorporate diverse external data during the pre-training phase, certain model architectures may be better suited for specific tasks than others. We will elaborate on this point later.

Detectors are also classifiers! We utilize YOLOv8 as the box and polygon proposal for building detection. However, we train it to simultaneously detect and classify objects. **Why classify if there is a classifier afterward?** In fact, since undamaged buildings constitute the majority of the data, the detector might deem them unnecessary to detect, especially if their features are not always obvious, making their impact negligible on the loss. To reinforce this, we cloned the repository from **Ultralytics Repo** and incorporated the **'pos-weight'** argument into the YOLO module to adjust the loss weighting by class. Remember, our objective is not to classify buildings but rather to minimize the likelihood of missing any buildings to the best of our ability. The accurate classification will be handled by classifiers.

Leveraging the given building's footprint: After using QGIS, we noticed certain buildings with unique shapes. This prompted us to employ an unsupervised model on building footprints and analyze the results. We discovered that with 4 clusters, we could reliably identify most commercial buildings (see [here](#)), the unsupervised model focuses on polygon characteristics like area, perimeter, compactness, and local density. This resolution addressed the challenge we encountered due to limited data for classifying residential and commercial buildings in a supervised manner. However, for the predictive phase to leverage this model effectively, it became apparent that having the polygon shapes of buildings, rather than just bounding boxes, was crucial. Hence, we opted for the polygon segmentation version of YOLO. Fortunately, external data sources with annotated building polygons were available for training the model. Ideally, annotating additional data from the provided dataset and fine-tuning YOLO would have been optimal, but time constraints prevented us from pursuing this option, especially given that this idea arose later in the process. Nevertheless, we capitalized on the detection of commercial buildings through unsupervised methods by cropping them and creating a new dataset specifically for commercial buildings, which we used to train a classifier

Tricky Data Augmentation: We applied data augmentation techniques to minority classes, including flipping, rotation, noise addition, and brightness adjustment. Additionally, we utilized corresponding pre-event images for annotated post images. For each annotated post-event image, we checked if the corresponding pre-event image was clear and changed the label of damaged buildings to undamaged ones. We also adjusted box boundaries slightly to account for the observed slight offset between pre-event and post-event satellite images.

Choosing External Data Wisely: To enhance the commercial building classifier, we incorporated external data consisting of industrial areas, parking lots with cars, and residential areas (uncropped, full images). However, by employing the DenseNet model, which can maintain level-based context in images, our aim is to enable the model to acquire features necessary for accurate commercial building detection. We classified severe damage and destruction from external data 1 (xView2 data) as damaged, treating minor damage as undamaged. We also included data on damaged buildings from the Turkey earthquake. Please refer to model_development.ipynb for dataset snippets and handling details, and check the ReadMe file for data sources.

